RAJALAKSHMI ENGINEERING COLLEGE

Arunchunai Nilesh S 251801026-2025-AIDS

A2

Started on	Monday, 31 August 2026, 2:02 PM
State	Finished
Completed on	Monday, 31 August 2026, 2:03 PM
Time taken	1 min 24 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement:

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

Output Format

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
int binarySearch(int arr[], int low, int high, int target)
{
    if (low > high)
        return -1;
    int mid = low + (high - low) / 2;
    if (arr[mid] == target)
        return mid;
    if (arr[mid] < target)
        return binarySearch(arr, mid + 1, high, target);
    return binarySearch(arr, low, mid - 1, target);
}
int findPair(int arr[], int n, int x, int index)
{
    if (index >= n)
        return -1;
    int complement = x - arr[index];
    int pos = binarySearch(arr, index + 1, n - 1, complement);
    if (pos != -1)
        return pos;
    return findPair(arr, n, x, index + 1);
}
int main()
{
    int n, x, i;
    scanf("%d", &n);
    int arr[n];
    for (i = 0; i < n; i++)
    {
        scanf("%d", &arr[i]);
    }
    scanf("%d", &x);
    int pos = findPair(arr, n, x, 0);
    if (pos != -1)
    {
        int firstIndex = pos;
        for (i = 0; i < pos; i++)
        {
            if (arr[i] + arr[pos] == x)
            {
                firstIndex = i;
                break;
            }
        }
        printf("%d\n", arr[firstIndex]);
        printf("%d\n", arr[pos]);
    }
    else
    {
        printf("No");
    }
    return 0;
```

	Input	Expected	Got	
✔	4 2 4 8 10 14	4 10	4 10	✔
✔	5 2 4 6 8 10 100	No	No	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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